

SYED MUHAMMAD HAMZAH RIZVI

Associate AI Solutions Architect

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PROFILE

AI engineer with 3+ years of customer-facing experience taking AI products from ambiguous requirements into reliable production deployments, across up to 15 concurrent projects spanning 10+ countries and thousands of operational endpoints. Designed and shipped a retrieval-augmented assistant now serving 300+ customer accounts weekly, using hybrid retrieval, reranking and grounding checks to suppress hallucinated answers. Combines applied LLM and conversational-AI work — RAG, semantic and knowledge search, prompt design and grounding — with the API integration, solution design and stakeholder skills needed to prototype, demonstrate and validate enterprise AI use cases. Translates business challenges into working proof-of-concepts and clear technical guidance for technical and non-technical audiences alike.

CORE SKILLS

AI / LLMs: RAG, semantic & knowledge search, conversational AI, prompt design, grounding/hallucination reduction, model evaluation, LLM APIs, self-hosted models.

Solution & Client Delivery: Requirements discovery, solution design, proof-of-concept & prototype building, technical workshops, demos, stakeholder management, technical documentation & enablement.

APIs & Integration: REST APIs, webhooks, authentication flows, JSON, API design & debugging, integrating AI with customer platforms, business systems and external data sources.

Programming & Tools: Python, C++, FastAPI, TensorFlow/Keras, ChromaDB, Ollama, Git; ChatGPT / Claude / Copilot for AI-assisted development.

EXPERIENCE

Technical Software Consultant — Innovative Technology, Manchester, UK

Mar 2023 – Present

- **Act as the primary technical advisor** on customer-facing deployments of AI-powered biometric and identity-verification solutions, owning delivery across engineering, product, commercial teams and end users.
- Run **discovery and requirements sessions** with customer engineering and business stakeholders, then design integration approaches and solution roadmaps aligned to their objectives.
- Build and troubleshoot **REST API integrations, webhooks and authentication flows** that connect AI products to customer platforms and external systems, moving projects from concept to live production rollout.
- Lead **technical workshops, solution reviews and product walkthroughs**, evaluating options with customers and preparing technical recommendations alongside commercial teams.
- Diagnose complex production issues across APIs, devices, mobile apps, firmware and networks, feeding findings back to product and engineering to improve the solution.
- Create **reusable technical assets** — integration guides, API documentation, diagrams and knowledge-transfer material — used by customers and internal teams to accelerate adoption.
- Supported rollouts across retail, gaming and compliance-focused environments spanning thousands of operational endpoints.

Automation Engineer — Tripack Films Ltd., Pakistan

Mar 2020 – Sep 2021

- Developed and supported industrial automation workflows, integrating control systems and sensors while debugging real-time hardware/software issues in production.

Software Consultant — Evolligence, Pakistan

Dec 2019 – Mar 2020

- Built and supported backend application logic using structured APIs and data-driven workflows, collaborating with development teams on delivery.

SELECTED AI PROJECTS

GroundedOps — Production RAG Support Assistant *Python, FastAPI, React, ChromaDB, Ollama, Docker, DeepSeek/OpenAI/Claude APIs*

- Deployed to production and now serving 300+ customer accounts weekly through a public embeddable widget and admin console; associated with a 40% reduction in first-line ticket volume and a 55% reduction in resolution time per internal Jira metrics since release (tracked alongside other concurrent initiatives, so not isolated to this tool alone).
- Built a retrieval-augmented QA assistant for technical documentation using hybrid retrieval (BM25 + dense embeddings), sigmoid-calibrated cross-encoder reranking and NLI-based grounding checks to reduce hallucination — directly analogous to virtual-agent and knowledge-search use cases.
- Replaced fragile follow-up heuristics with an LLM-based Rewrite-Retrieve-Read query-condensation flow and per-session conversation memory; backed by a 140+ test regression suite for fast, evidence-driven iteration.

Age Estimation & Demographic Analysis (MSc ML Project) *TensorFlow/Keras, EfficientNetB0, VGG16*

- Trained a hierarchical multi-task EfficientNetB0 model on 274k+ facial images (WIKI/IMDB, fine-tuned on UTKFace) with end-to-end data pipelines and evaluation tooling, achieving 87.5% gender accuracy, 88.4% coarse-age accuracy, 92.6% near-group (± 1) accuracy and 8.52-year MAE.

EDUCATION

MSc Robotics and Automation (Minor in AI) — University of Salford, UK

Sep 2021 – Nov 2022

Graduated with Distinction. Focus: AI in biometrics, mobile robotics and automation. Dissertation: facial age & gender estimation using CNNs.

BEng Electrical Engineering — NUST, Pakistan

Sep 2015 – Aug 2019

Focus: computer vision, machine learning and robotics.